```
shubham@k8s-containerd-master:~$ kubectl get nodes -o wide
NAME                    STATUS   ROLES                  AGE     VERSION   INTERNAL-IP      EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION      CONTAINER-RUNTIME
k8s-containerd-master   Ready    control-plane,master   3m17s   v1.23.6   192.168.122.11   <none>        Ubuntu 20.04.4 LTS   5.4.0-109-generic   containerd://1.6.2
k8s-containerd-worker   Ready    <none>                 99s     v1.23.6   192.168.122.12   <none>        Ubuntu 20.04.4 LTS   5.4.0-109-generic   containerd://1.6.2
shubham@k8s-containerd-master:~$
```

Figure 10: Kubernetes nodes with containerd runtime

```
shubham@k8s-containerd-master:~$ kubectl get pods --all-namespaces -o wide
NAMESPACE     NAME                                            READY   STATUS    RESTARTS   AGE     IP               NODE                    NOMINATED NODE   READINESS GA
TES
kube-system   coredns-64897985d-8q544                         1/1     Running   0          2m22s   10.244.0.2       k8s-containerd-master   <none>           <none>
kube-system   coredns-64897985d-j4jb6                         1/1     Running   0          2m22s   10.244.0.3       k8s-containerd-master   <none>           <none>
kube-system   etcd-k8s-containerd-master                      1/1     Running   0          2m43s   192.168.122.11   k8s-containerd-master   <none>           <none>
kube-system   kube-apiserver-k8s-containerd-master            1/1     Running   0          2m49s   192.168.122.11   k8s-containerd-master   <none>           <none>
kube-system   kube-controller-manager-k8s-containerd-master   1/1     Running   0          2m50s   192.168.122.11   k8s-containerd-master   <none>           <none>
kube-system   kube-flannel-ds-jrwlt                           1/1     Running   0          93s     192.168.122.11   k8s-containerd-master   <none>           <none>
kube-system   kube-flannel-ds-lgvz5                           1/1     Running   0          75s     192.168.122.12   k8s-containerd-worker   <none>           <none>
kube-system   kube-proxy-h8tmv                                1/1     Running   0          75s     192.168.122.12   k8s-containerd-worker   <none>           <none>
kube-system   kube-proxy-qjcmg                                1/1     Running   0          2m22s   192.168.122.11   k8s-containerd-master   <none>           <none>
kube-system   kube-scheduler-k8s-containerd-master            1/1     Running   0          2m42s   192.168.122.11   k8s-containerd-master   <none>           <none>
shubham@k8s-containerd-master:~$
```

Figure 11: All the pods in the Kubernetes cluster and their status nodes with containerd runtime

can be found at *https://github.com/shubhamaggarwal890/nginx-vod/blob/master/kubernetes-containerd.md.*

Figure 10 shows all the nodes attached to the Kubernetes cluster, where we can see our installed master and worker nodes. The CONTAINER-RUNTIME column shows the installed container runtime along with its version. Figure 11 showcases all the pods attached to the name spaces. We can see certain pods running on the master node and on the worker node under the NODE column.

Kubernetes orchestrates the running containers and helps them scale up and down based on the need of the hour. The application can be easily patched with the latest changes with a simple rollout from Kubernetes, while also handling rollback in case that is needed. The self-healing feature of Kubernetes helps restart the pods if they crash at any time.

The application was successfully installed over the Kubernetes cluster. There is a plethora of ways in which a Kubernetes cluster can be set up. For the purpose of development, an easy installation of minikube is fine, but for a production-based system, one should go for the installation of master and worker nodes. The choice of container runtime lies with the users; they can choose from a wide variety of container runtimes such as containerd, CRI-O, and Docker.

One can easily install the Kubernetes cluster as shown above, but where to install it is the question. Today, various cloud services offer a Kubernetes engine, but none of these solutions are open source. What if some organisation wants to host an in-house solution for the orchestration of containers? Here, Infrastructure as a Service is a move in the right direction, where we can set up a cloud infrastructure that caters to the computation, networking, and storage requirements. With the advent of open source cloud computing infrastructure, an organisation can easily enhance its computing resources in case of traffic growth, giving it an edge over others that are still embracing traditional deployments. END

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